PRISM | Google Cloud edition

Original fleet workflow, grounded operational assistance

- Manufacturing | Intelligent Operations & Industrial Efficiency

- For plant supervisors and reliability engineers: turn fragmented alarms and sensor readings into a traceable incident brief.

- Original 22-asset PRISM simulator and six-tab UI. No equipment control or measured production savings.

The problem and the narrow wedge

One upstream fault should not become twenty tickets

- Fleet triage for plant equipment and its power chain, with chronological telemetry replacement.
- Group downstream supply symptoms without hiding independent bearing, pressure or battery risks.
- Product hypothesis: trained evidence, topology and a cited engineer handoff reduce gathering effort.
- Validate this hypothesis with an approved pilot; no invented customer or market statistics.

A complete, reviewable workflow

Monitor -> Diagnose -> Decide -> Act -> Verify

- Monitor: original history-trained robust-z + Isolation Forest and Markov event-pattern tools.
- Diagnose: original topology correlation, plus Vertex AI hypotheses and validated SOP passages.
- Decide/Act: original reversibility/confidence policy; automatic actions only change the simulator.
- Verify: original subsequent-sample verifier and three-sample grace period. Uploaded data is observation-only.
- Exact offline parity: all 96 snapshots, case fields, model outputs and counterfactuals match PRISM.

Google Cloud architecture

Real services, not decorative integrations

- Cloud Run: deployed application, scale-to-zero, one-instance demo limit.
- Gemini on Vertex AI: structured incident reasoning and work-order drafting.
- Cloud Storage: private SOP and synthetic scenario documents.
- BigQuery: partitioned audit table, 30-day retention, bounded history query.
- Dedicated runtime service account; no API key embedded in the cloud service.

Validation and honest limitations

Make failures visible; do not manufacture success

- Regression tests exercise approval, invalid output, bad citations, missing evidence, reopened cases, storage failure and rejected audit rows.

- Recorded live smoke test: 4/4 accepted Gemini diagnoses. This is not accuracy on a labeled production dataset.

- Quote checks establish textual provenance, not proof of every causal claim. Model confidence is self-reported.

- Built-in assets/SOPs are synthetic. Original value fields are assumed counterfactuals, not realized savings.

Impact, feasibility and the next step

Start with one approved workflow at one plant

- Pilot measures: manual-vs-assisted triage time, engineer acceptance, unsupported claims and appropriate escalation.

- Use approved site procedures, qualified industrial review, tenant isolation and authenticated access before any production pilot.

- Scale documents and audit across sites; add real monitoring ingestion only after validation.

- Future roadmap only: Pub/Sub, Dataflow and hosted predictive workloads.

- Submission: deployed demo + public source + this PDF + video strictly under three minutes.